Task 1:

Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

Answer:

Top 3 Protocols Observed in Wireshark

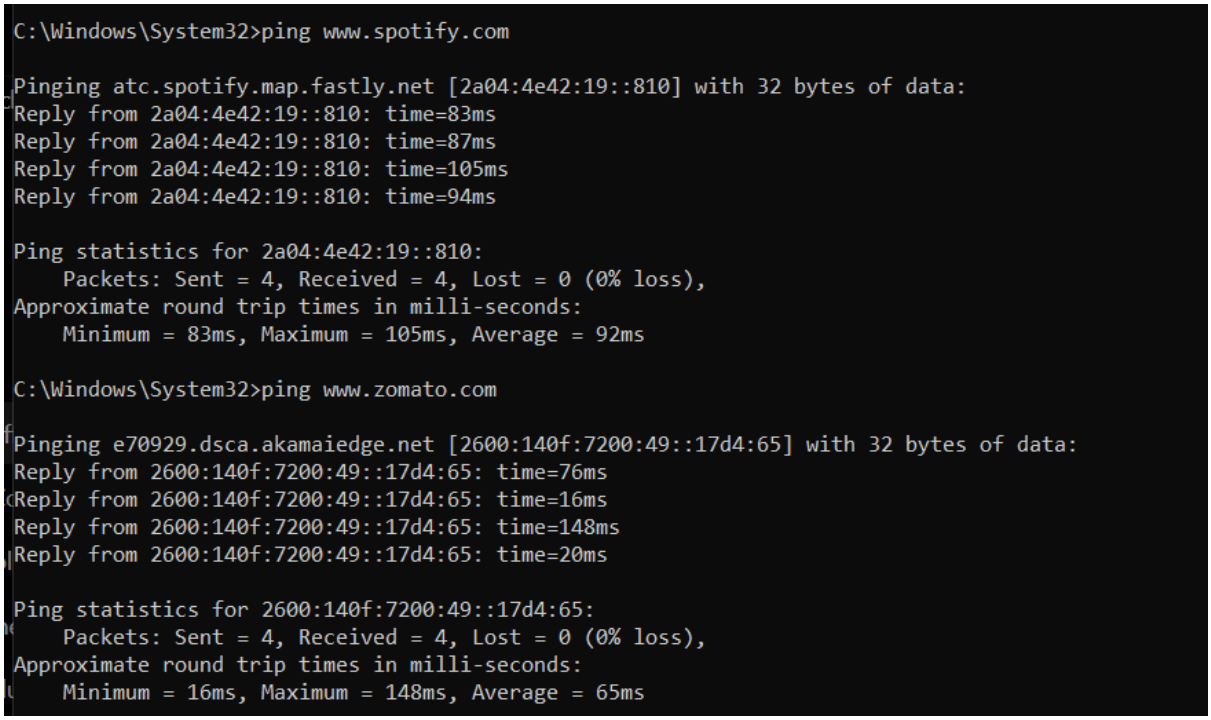
Protocol	Full Form	Purpose
UDP	User Datagram Protocol	Provides fast data transmission without guaranteeing delivery.
TCP	Transmission Control Protocol	Provides reliable and ordered data transmission.
HTTP	Hypertext Transfer Protocol	Transfers web page data between a browser and a web server.

Result: Wireshark was used to capture network traffic while browsing the internet. The observed protocols were UDP, TCP, and HTTP.

Task 2:

Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

Answer:

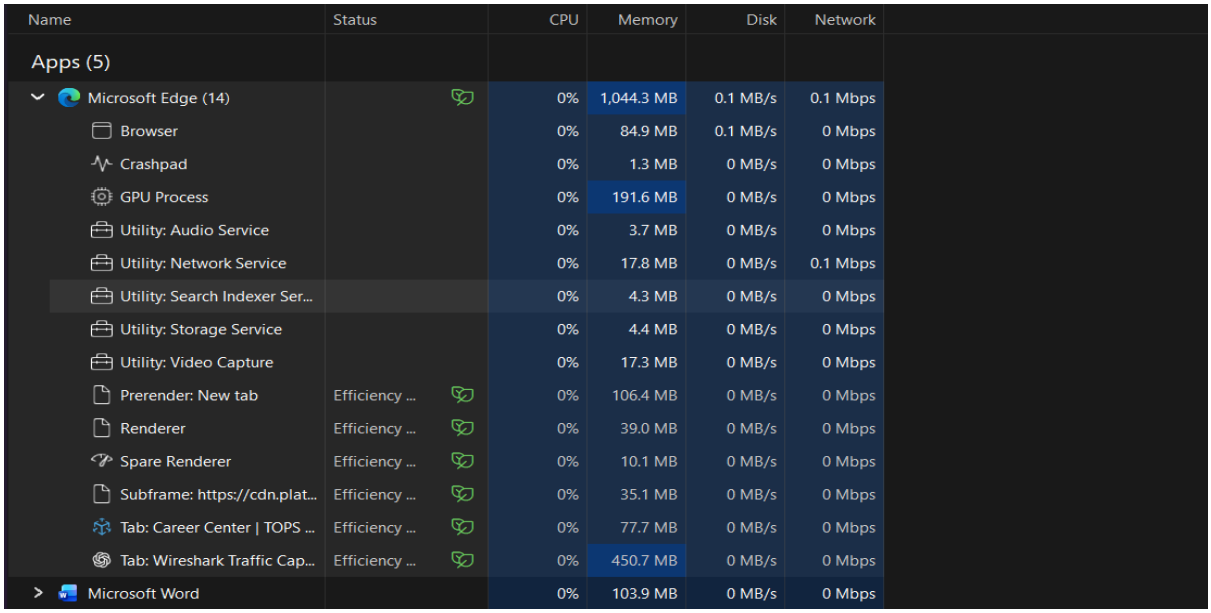


Zomato site is give faste response.

Task 3:

Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.

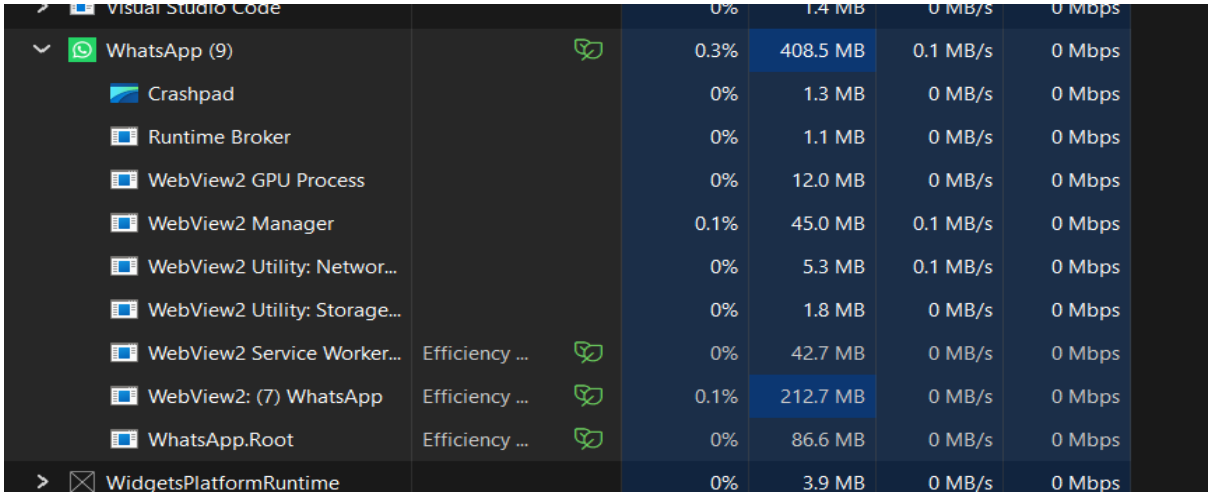
Answer:



Name	Status	CPU	Memory	Disk	Network
Apps (5)					
Microsoft Edge (14)		0%	1,044.3 MB	0.1 MB/s	0.1 Mbps
Browser		0%	84.9 MB	0.1 MB/s	0 Mbps
Crashpad		0%	1.3 MB	0 MB/s	0 Mbps
GPU Process		0%	191.6 MB	0 MB/s	0 Mbps
Utility: Audio Service		0%	3.7 MB	0 MB/s	0 Mbps
Utility: Network Service		0%	17.8 MB	0 MB/s	0.1 Mbps
Utility: Search Indexer Ser...		0%	4.3 MB	0 MB/s	0 Mbps
Utility: Storage Service		0%	4.4 MB	0 MB/s	0 Mbps
Utility: Video Capture		0%	17.3 MB	0 MB/s	0 Mbps
Prerender: New tab	Efficiency ...	0%	106.4 MB	0 MB/s	0 Mbps
Renderer	Efficiency ...	0%	39.0 MB	0 MB/s	0 Mbps
Spare Renderer	Efficiency ...	0%	10.1 MB	0 MB/s	0 Mbps
Subframe: https://cdn.plat...	Efficiency ...	0%	35.1 MB	0 MB/s	0 Mbps
Tab: Career Center | TOPS ...	Efficiency ...	0%	77.7 MB	0 MB/s	0 Mbps
Tab: Wireshark Traffic Cap...	Efficiency ...	0%	450.7 MB	0 MB/s	0 Mbps
Microsoft Word		0%	103.9 MB	0 MB/s	0 Mbps



NVIDIA Container (2)	0.3%	25.5 MB	0.3 MB/s	0 Mbps
OMEN Gaming Hub (2)	1.0%	114.6 MB	0.1 MB/s	0 Mbps
OmenCap.exe	0%	1.1 MB	0 MB/s	0 Mbps
OneDrive	0%	1.9 MB	0 MB/s	0 Mbps
Opera Browser Assistant (32 bi...	0%	5.9 MB	0 MB/s	0 Mbps
Opera Browser Assistant (...	0%	4.4 MB	0 MB/s	0 Mbps
Opera Browser Assistant (...	0%	1.5 MB	0 MB/s	0 Mbps
Phone Link	0%	41.0 MB	0 MB/s	0 Mbps



Visual Studio Code		0%	1.4 MB	0 MB/s	0 Mbps
WhatsApp (9)		0.3%	408.5 MB	0.1 MB/s	0 Mbps
Crashpad		0%	1.3 MB	0 MB/s	0 Mbps
Runtime Broker		0%	1.1 MB	0 MB/s	0 Mbps
WebView2 GPU Process		0%	12.0 MB	0 MB/s	0 Mbps
WebView2 Manager		0.1%	45.0 MB	0.1 MB/s	0 Mbps
WebView2 Utility: Networ...		0%	5.3 MB	0.1 MB/s	0 Mbps
WebView2 Utility: Storage...		0%	1.8 MB	0 MB/s	0 Mbps
WebView2 Service Worker...	Efficiency ...	0%	42.7 MB	0 MB/s	0 Mbps
WebView2: (7) WhatsApp	Efficiency ...	0.1%	212.7 MB	0 MB/s	0 Mbps
WhatsApp.Root	Efficiency ...	0%	86.6 MB	0 MB/s	0 Mbps
WidgetsPlatformRuntime		0%	3.9 MB	0 MB/s	0 Mbps

Task 4:

use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

Answer:

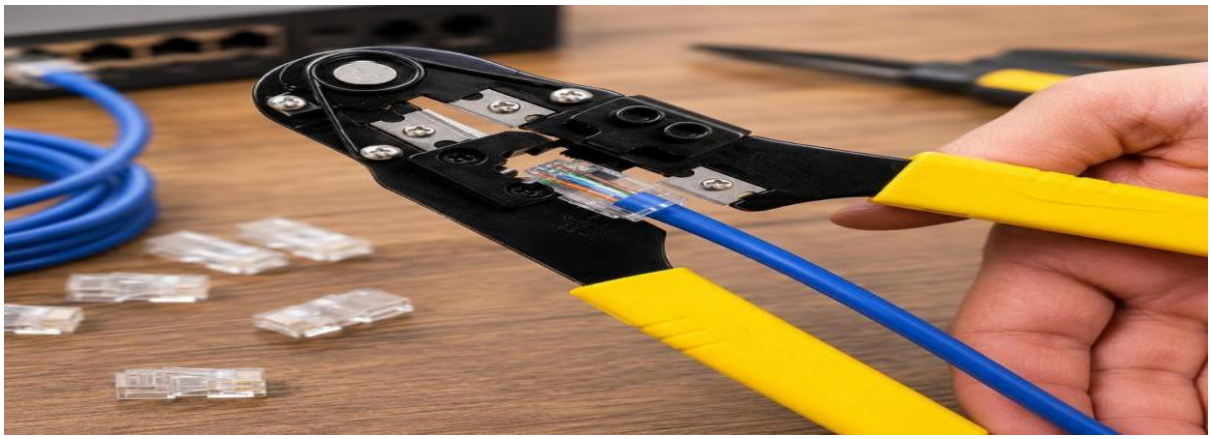
Cable Tester



A cable tester is a hardware tool used to check whether an Ethernet or LAN cable is working correctly. It helps identify problems such as broken wires, incorrect wire connections, short circuits, and open circuits.

I would use a cable tester when a computer cannot connect to the network and I suspect that the Ethernet cable is damaged or incorrectly connected.

Crimping Tool



A crimping tool is used to attach an RJ45 connector to the end of an Ethernet cable. It presses the connector firmly onto the cable wires to create a proper connection. It is commonly used when making new LAN cables or repairing cables with damaged connectors.

When would you use it? I would use a crimping tool when preparing a new Ethernet cable or replacing a damaged RJ45 connector so that the cable can connect a computer to a router or switch.

3. Digital Multimeter



A digital multimeter is an electronic measuring instrument used to measure electrical voltage, resistance, and continuity. It helps troubleshoot electrical problems in network equipment, such as power adapters, switches, and routers. It should be used according to the equipment's electrical specifications and safety instructions.

When would you use it? I would use a digital multimeter when a router or network switch does not turn on and I need to check whether its power adapter is supplying the correct voltage.